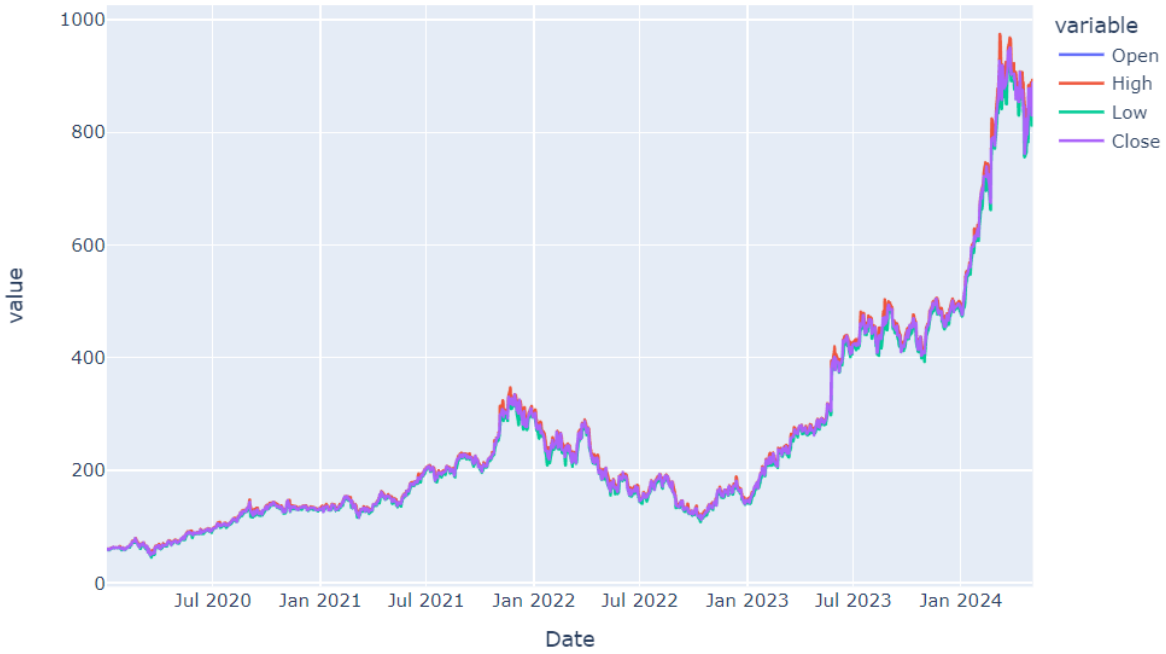
Assignment 1: Part B

Dataset Features:

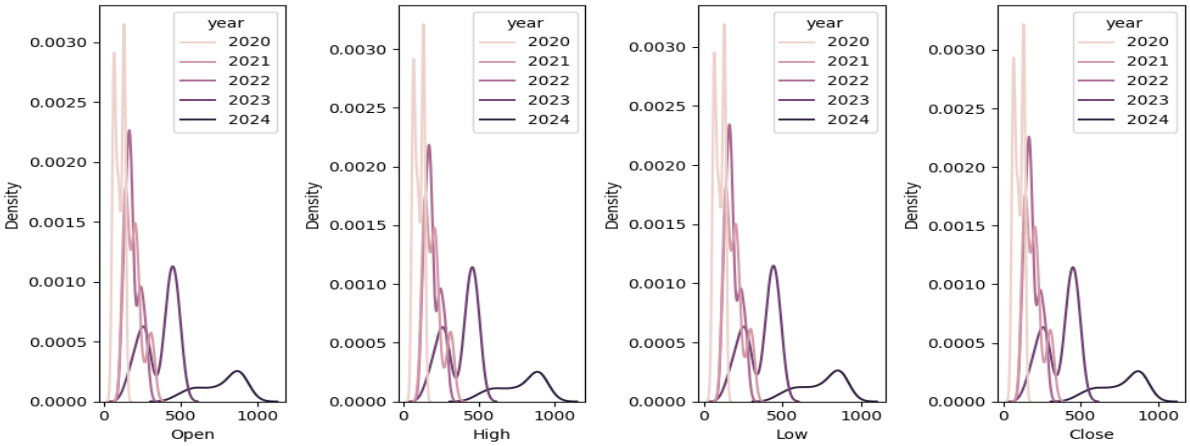
- **Daily Stock Price Data**: This includes daily opening, closing, high, and low prices of NVIDIA stock. These variables are essential for analyzing daily price movements and identifying trends or patterns.
- **Trading Volume**: Daily trading volume data, which measures the total number of shares traded each day. It provides insights into market activity, liquidity, and potential investor sentiment.
- **Event Dataset**: A collection of events categorized by type. Each event is associated with specific dates, allowing for alignment with daily stock data to analyze price movements around these events.
- **Event Types**: Classification of events based on their nature. This classification helps in understanding how different types of events impact NVIDIA's stock price differently.
- **Historical Data Period**: The timeframe of historical data available for analysis is crucial for understanding trends over time and assessing the consistency of relationships between volume, price movements, and events.

Initial Data Analysis and Visualization:



The overall trend of the stock is upward, with periods of stabilization and volatility. The increasing values suggest the stock has gained popularity and possibly reflects positive performance or investor sentiment and the volatility, particularly in the later stages, could be attributed to market events, investor speculation, or company-specific news. To simplify the task, the idea was to investigate the data and patterns by separating the data into yearly segments.

Data Distribution among years:

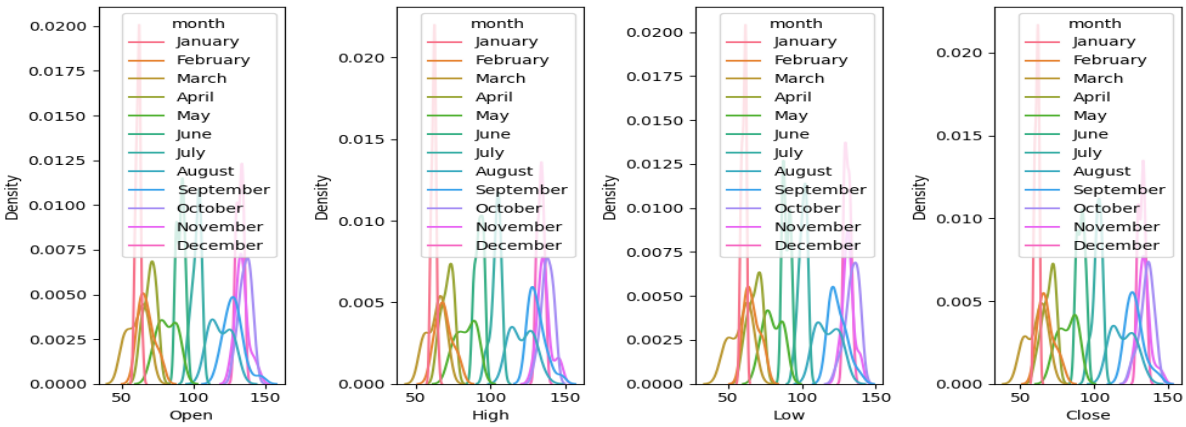


This graph shows the kernel density estimation (KDE) plots of the Open, High, Low, and Close values of a stock or financial asset, segmented by year (2020-2024).

- 2020: The density peaks around a lower value, indicating that the stock prices were generally lower during this year.
- 2021: The density is slightly shifted higher compared to 2020, suggesting an increase in stock prices.
- 2022: The distribution continues to shift to the right, indicating further increases in stock prices.
- 2023: The peaks are broader and more to the right, implying a wider range of higher stock prices.
- 2024: There is a noticeable peak at higher values, suggesting a significant increase in stock prices compared to previous years.

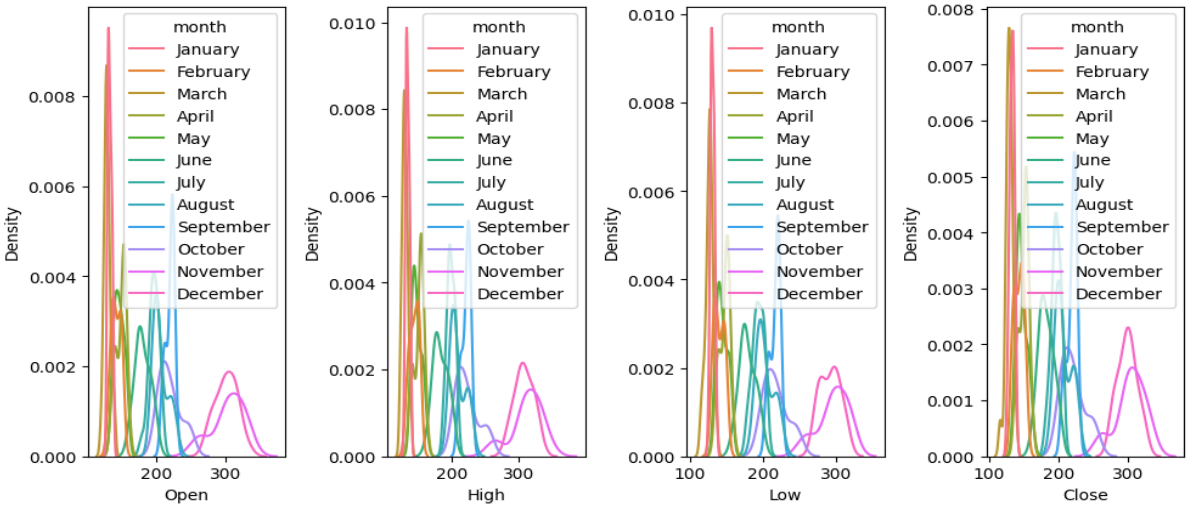
Monthly stocks per month for each year:

2020



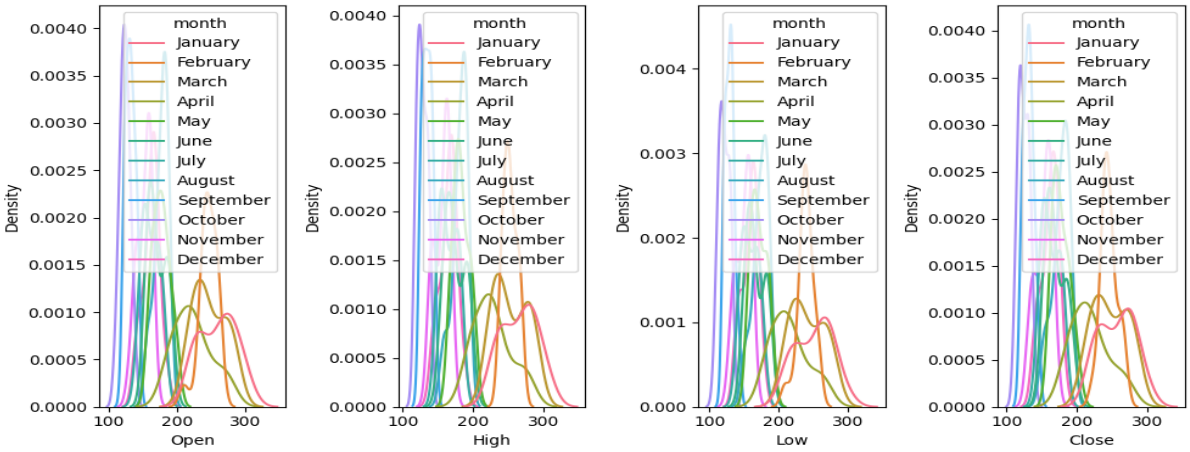
January to December each month's density plot shows a distribution of stock prices within the range of approximately 50 to 150. There is significant overlap between the months, suggesting that while there are some variations, the stock prices for most months fall within a similar range.

2021



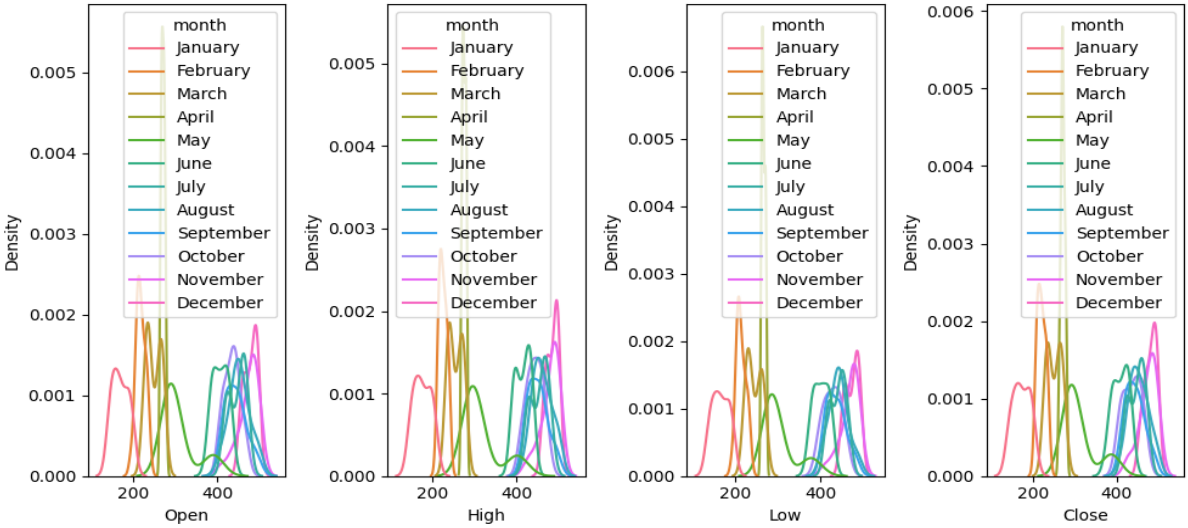
From January to December the density plots now show distributions of stock prices within the range of approximately 100 to 350. Similar to the initial stock prices, there are variations in the peak densities for different months, indicating seasonal trends and there is still significant overlap between the months, but the overall range of stock prices has shifted upwards.

2022



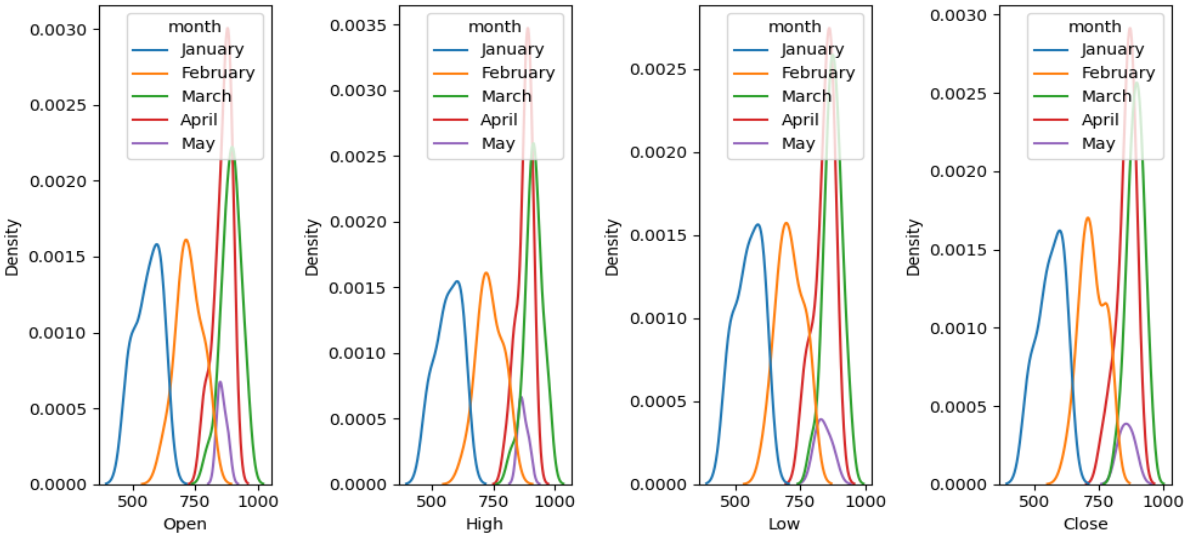
From January to December the density plots now show distributions of stock prices within the range of approximately 100 to 300.There is a general increase in stock prices from March to December. The density distributions from March to December are more spread out compared to January and February, suggesting higher volatility or a wider range of stock prices.

2023



From January to December the density plots now show distributions of stock prices within the range of approximately 200 to 400. January and February show less spread in the distribution, indicating more stable or less volatile stock prices.

2024



From January to December the density plots now show distributions of stock prices within the range of approximately 500 to 1000. From February to May there are more concentrated peaks, suggesting less volatility and more stable stock prices.

1. Seasonal Variation: The above sets of density plots show that there are seasonal variations in stock prices, with certain months exhibiting higher or lower prices consistently.
2. Volatility: The later time period shows increased volatility, reflecting a more significant variation in stock prices.

3. Overlap: Despite the seasonal variations, there is a consistent overlap between the months, meaning that while there are monthly trends, the stock prices do not drastically differ month-to-month.
4. Unimodal Distributions: Most density plots are unimodal, suggesting that each month's stock prices are centered around a single value with variability around it.

These observations provide a contextual backdrop for understanding how trading volume correlates with daily price movements and how specific events influence NVIDIA's stock price dynamics. They main highlight is that there is a noticeable shift in the KDE peaks towards higher stock prices as the year progresses. These plot shows the interplay between market behaviour, seasonal trends, volatility, and event-driven catalysts, offering valuable insights for our research seeking to navigate and capitalize on fluctuations in NVIDIA's stock market performance.

Question Refinement and Additional Data Requirement:

How do trading volume and specific events, categorized by type, correlate with daily price movements (open, close, high, low) of NVIDIA stock, and what insights can be drawn from these relationships regarding market behaviour, investor sentiment, and strategic decision-making?

Additional data is required of the NVIDIA events such as Earnings Releases, Investor Conferences, Product Launches, Acquisitions and Mergers, Legal and Regulatory Developments, Industry Events and Keynotes, etc. and then mapping it to the changes in the stock prices due to the same (in order to satisfy the 4V's of Big Data Analysis).

References:

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4. www.marketwatch.com. (n.d.). NVIDIA stock data. [online] Available at: https://www.marketwatch.com/investing/stock/nvda/download-data [Accessed 29 June 2024].
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